

CS3 Rubric – Predicting Housing Prices Case Study

DS 4002 – Spring 2026 - Instructors: Javier Rasero; TA: Ethan Meidinger

Due: May 4, for comments; 2:59 pm, May 6 (Hardest deadline possible! -- any submission after this will not count)

Submission format: Upload link to GitHub repo to Canvas and a PDF Presentation

Individual Assignment

Why am I doing this? This assignment places you in the role of a data scientist advising local policymakers on housing affordability in Charlottesville, Virginia. Your task is to determine whether uncertainty expressed in community meeting minutes can predict changes in the local housing market. You will learn how to extract signals from unstructured text, merge them with quantitative market data, and model their relationship over time — communicating your decisions clearly for a broad audience.

- Course Learning Objective: Use analysis and critical thinking skills
- Course Learning Objective: Prepare findings for presentation to your peers.

What am I going to do? First, read the Hook Document to understand the scenario and goal of the assignment. Then, review the included resources on NLP text analysis, time series methods, and housing market data. Use the README and resources in the GitHub repo to load the CAAR market reports and HAC meeting minutes, extract uncertainty signals from the text, merge and clean the data, and build a time series regression model. Lastly, prepare a presentation that walks the audience through your workflow, decisions, findings, and recommendations.

Deliverables include:

- GitHub Repository – contains all annotated code, figures, and resources so your results can be reproduced
- Presentation – a slide deck guiding the audience through your analysis

Tips for success:

- Do not overthink it. A clear presentation of fundamentals is more valuable than a complicated approach.
- Talk to instructors and classmates. You may get feedback and ask questions.
- Explain your reasoning. This is as important as your model performance.

How will I know I have succeeded? You will meet expectations on CS3 Create Case Study when you follow the criteria in the rubric below.

Important: Please begin by cloning the GitHub repository. Once downloaded, read the Hook Document to understand the scenario and purpose of the case study. Then review the README.md, which provides essential information about the project structure, the files included, how to run the workflow, and common troubleshooting tips. This document will guide you through reproducing the analysis and completing the assignment successfully.

Formatting	• Repository – A GitHub repo (and cloud storage folder if necessary) containing all materials 3 - o Submit a link to the repo - o Everything is contained in the repo or linked to it if appropriate. - o Contents ▪ README.md ▪ LICENSE.md ▪ SCRIPTS folder ▪ DATA folder ▪ OUTPUT folder ▪ Presentation slides in PDF format - o Use PDF format when possible - o For code and data products, use the appropriate format for whatever it is • Presentation - o Should contain a 10 to 12-slide deck in PDF format
GitHub Repository	• <u>Goal</u>: This repo is your professional, reproducible record of analysis. Someone should be able to download your repo and fully recreate your results. • README.md - o Explain how to use the repo - o Use section headers (##) - o Include software requirements, environment information, and instructions for reproducing results - o Provide a map of the repository structure • LICENSE.md - o Choose an appropriate license (MIT is usually acceptable) • SCRIPTS folder - o Include all six numbered notebooks used in the analysis - o Name scripts in the order needed to reproduce results (e.g. 01_scrape_meeting_mins.ipynb) - o Include header comments describing purpose, inputs, and outputs - o Include clear comments throughout explaining each major step • DATA folder - o Include the raw CAAR PDF reports and HAC meeting minutes .txt files - o Include any cleaned or merged CSV files produced during the pipeline

	- o Code in SCRIPTS must be able to reproduce the final data from the initial raw files • OUTPUT folder - o Include all figures used in your presentation (time series plot, correlation heatmap, scatter plot) - o Use informative filenames (e.g., uncertainty_vs_price_timeseries.png) • Copy of the presentation - o Include the finalized slide deck in PDF format
Presentation	<u>Goal:</u> This presentation is the primary method of communicating your results. It must guide your audience through your analysis, from problem framing to conclusions. Your slide deck must include the following: • Title and Outline (1 slide) - o State your main takeaway in headline format - o Give an overview of the presentation flow • Motivation, Context, Research Question, and Modeling Approach (1 to 2 slides) - o Explain why housing affordability and community uncertainty matter in Charlottesville - o Restate the research question - o Summarize the pipeline you chose (text extraction → merge → clean → model) • Data Explanation and Acquisition (1 slide) - o Summarize both data sources: CAAR monthly reports and HAC meeting minutes - o Describe key variables, date range, and any gaps in the data (e.g., missing August minutes) - o Summarize how each dataset was obtained and any limitations • Analysis Plan and Justification (1 slide) - o Present your pipeline diagram from text scraping to regression output - o Explain uncertainty signal extraction and how it was operationalized - o Justify your choice of time series regression model • Tricky Analysis Decision (1 slide) - o Describe a judgment call (e.g., how uncertainty words were selected, how missing months were handled, lag structure choice) - o Explain why it mattered and the impact on results • Bias and Uncertainty Validation (1 slide) - o Discuss limitations of the dataset (small sample size, subjective word lists, PDF extraction errors)

	- o Describe how you assessed model uncertainty and whether findings generalize • Results and Conclusions (1 slide) - o Present your regression output clearly (coefficients, p-values, R-squared) - o Include your key visualizations (time series, heatmap, scatter) - o Address your research question directly • Next Steps (1 slide) - o Suggest improvements to the pipeline - o Discuss extensions or future directions (e.g., more months of data, other cities, additional text sources) • References and Acknowledgements (1 slide) - o Cite datasets, libraries, articles, and any resources used
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Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from [Streifer & Palmer \(2020\)](#).